E7 Solving trigonometric equations

Name: ______________________

Class: ______________________

Date: ______________________

Time: **384 minutes**

Marks: **318 marks**

Comments:

Q1.

Solve the equation $\tan^2 2\theta - 3 = 0$ giving all the solutions for $0° \le \theta \le 360°$

(Total 4 marks)

Q2.

Jessica, a maths student, is asked by her teacher to solve the equation $\tan x = \sin x$, giving all solutions in the range $0° \le x \le 360°$

The steps of Jessica's working are shown below.

$$\tan x = \sin x$$

Step 1	$\Rightarrow$	$\dfrac{\sin x}{\cos x} = \sin x$	Write $\tan x$ as $\dfrac{\sin x}{\cos x}$
Step 2	$\Rightarrow$	$\sin x = \sin x \cos x$	Multiply by $\cos x$
Step 3	$\Rightarrow$	$1 = \cos x$	Cancel $\sin x$
	$\Rightarrow$	$x = 0°$ or $360°$	

The teacher tells Jessica that she has not found all the solutions because of a mistake.

Explain why Jessica's method is not correct.

(Total 2 marks)

Q3.

Solve the equation

$$\sin\theta\tan\theta + 2\sin\theta = 3\cos\theta \quad \text{where } \cos\theta \ne 0$$

Give **all** values of θ to the nearest degree in the interval $0° < \theta < 180°$
Fully justify your answer.

(Total 5 marks)

Q4.

(a) Write down the two solutions of the equation $\tan(x + 30°) = \tan 79°$ in the interval $0° \le x \le 360°$.

(2)

(b) Describe a single geometrical transformation that maps the graph of $y = \tan x$ onto the graph of $y = \tan(x + 30°)$.

(2)

(c) (i) Given that $5 + \sin^2 \theta = (5 + 3\cos\theta)\cos\theta$, show that $\cos\theta = \frac{3}{4}$.

(5)

(ii) **Hence** solve the equation $5 + \sin^2 2x = (5 + 3\cos 2x)\cos 2x$ in the interval $0 < x < 2\pi$, giving your values of x in radians to three significant figures.

(3)

(Total 12 marks)

Q5.

(a) Show that

$$\frac{\sec^2 x}{(\sec x + 1)(\sec x - 1)}$$

can be written as $\operatorname{cosec}^2 x$.

(3)

(b) Hence solve the equation

$$\frac{\sec^2 x}{(\sec x + 1)(\sec x - 1)} = \operatorname{cosec} x + 3$$

giving the values of x to the nearest degree in the interval $-180° < x < 180°$.

(6)

(c) Hence solve the equation

$$\frac{\sec^2(2\theta - 60°)}{(\sec(2\theta - 60°) + 1)(\sec(2\theta - 60°) - 1)} = \operatorname{cosec}(2\theta - 60°) + 3$$

giving the values of θ to the nearest degree in the interval $0° < \theta < 90°$.

(2)

(Total 11 marks)

Q6.

(a) Express $3\cos x + 2\sin x$ in the form $R\cos(x - \alpha)$, where $R > 0$ and $0° < \alpha < 90°$, giving your value of α to the nearest 0.1°.

(3)

(ii) Hence find the minimum value of $3\cos x + 2\sin x$ and the value of x in the interval
$0° < x < 360°$ where the minimum occurs. Give your value of x to the nearest 0.1°.

(3)

(b) (i) Show that $\cot x - \sin 2x = \cot x \cos 2x$ for $0° < x < 180°$.

(3)

(ii) Hence, or otherwise, solve the equation

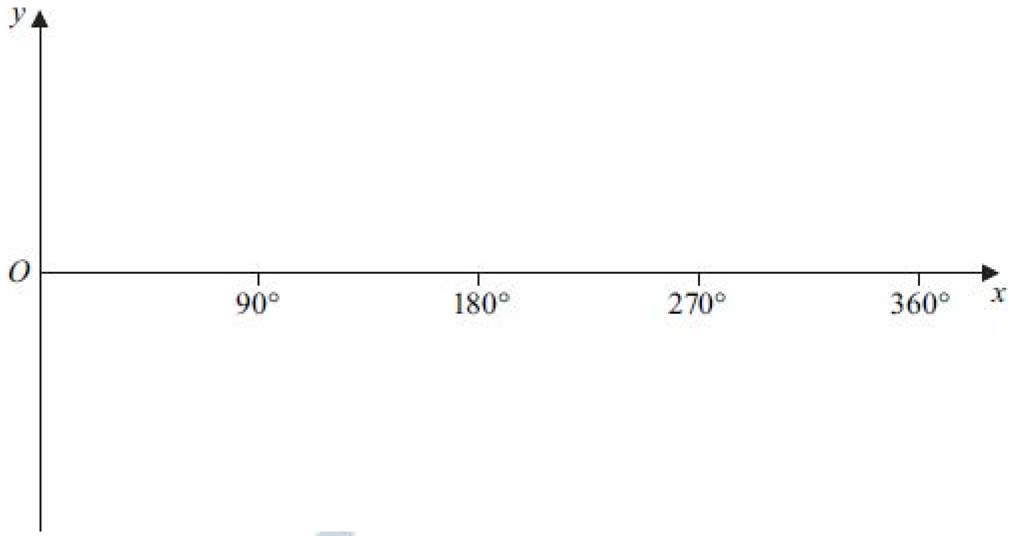
$$\cot x - \sin 2x = 0$$

in the interval $0° < x < 180°$.

(3)
(Total 12 marks)

Q7.

(a) (i) On the axes given below, sketch the graph of $y = \tan x$ for $0° \leq x \leq 360°$.



(3)

(ii) Solve the equation $\tan x = -1$, giving all values of x in the interval $0° \leq x \leq 360°$.

(2)

(b) (i) Given that $6 \tan \theta \sin \theta = 5$, show that $6 \cos^2 \theta + 5 \cos \theta - 6 = 0$.

(3)

(ii) **Hence** solve the equation $6 \tan 3x \sin 3x = 5$, giving all values of x to the nearest degree in the interval $0° \leq x \leq 180°$.

(6)
(Total 14 marks)

Q8.

By forming and solving a quadratic equation, solve the equation

$$8 \sec x - 2 \sec^2 x = \tan^2 x - 2$$

in the interval $0 < x < 2\pi$, giving the values of x in radians to three significant figures.

(Total 7 marks)

Q9.

The polynomial f(x) is defined by f(x)= $4x^3 - 11x - 3$.

(a) Use the Factor Theorem to show that $(2x + 3)$ is a factor of f(x).

(2)

(b) Write f(x) in the form $(2x + 3)(ax^2 + bx + c)$, where a, b and c are integers.

(2)

(c) (i) Show that the equation $2 \cos 2\theta \sin \theta + 9 \sin \theta + 3 = 0$ can be written as $4x^3 - 11x - 3 = 0$, where $x = \sin \theta$.

(3)

(ii) Hence find all solutions of the equation $2 \cos 2\theta \sin \theta + 9 \sin \theta + 3 = 0$ in the interval $0° < \theta < 360°$, giving your solutions to the nearest degree.

(4)

(Total 11 marks)

Q10.

(a) Given that $2 \sin \theta = 7 \cos \theta$, find the value of $\tan \theta$.

(2)

(b) (i) Use an appropriate identity to show that the equation

$$6 \sin^2 x = 4 + \cos x$$

can be written as

$$6 \cos^2 x + \cos x - 2 = 0$$

(2)

(ii) Hence solve the equation $6 \sin^2 x = 4 + \cos x$ in the interval $0° < x < 360°$, giving your answers to the nearest degree.

(6)

(Total 10 marks)

Q11.

(a) By using a suitable trigonometrical identity, solve the equation

$$\tan^2 \theta = 3(3 - \sec \theta)$$

giving all solutions to the nearest 0.1° in the interval $0° < \theta < 360°$.

(6)

(b) Hence solve the equation

$$\tan^2(4x - 10°) = 3[3 - \sec(4x - 10°)]$$

giving all solutions to the nearest 0.1° in the interval $0° < x < 90°$.

(3)
(Total 9 marks)

Q12.

(a) Use the Factor Theorem to show that $4x - 3$ is a factor of

$$16x^3 + 11x - 15$$

(2)

(b) Given that $x = \cos\theta$, show that the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

can be written in the form

$$16x^3 + 11x - 15 = 0$$

(4)

(c) Hence show that the only solutions of the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

are given by $\cos\theta = \frac{3}{4}$.

(4)
(Total 10 marks)

Q13.

It is given that $(\tan\theta + 1)(\sin^2\theta - 3\cos^2\theta) = 0$.

Find the possible values of $\tan\theta$.

(Total 4 marks)

Q14.

(a) Show that the equation

$$\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta} = 32$$

can be written in the form

$$\mathrm{cosec}^2\,\theta = 16$$

(4)

(b) Hence, or otherwise, solve the equation

$$\frac{1}{1+\cos(2x-0.6)}+\frac{1}{1-\cos(2x-0.6)} = 32$$

giving all values of x in radians to two decimal places in the interval $0 < x < \pi$.

(5)

(Total 9 marks)

Q15.

(a) Express $\sin x - 3\cos x$ in the form $R\sin(x - \alpha)$, where $R > 0$ and $0° < \alpha < 90°$, giving your value of α to the nearest 0.1°.

(3)

(b) Hence find the values of x in the interval $0° < x < 360°$ for which

$$\sin x - 3\cos x + 2 = 0$$

giving your values of x to the nearest degree.

(4)

(Total 7 marks)

Q16.

(a) Solve the equation $\tan x = -3$ in the interval $0° \le x \le 360°$ giving your answers to the nearest degree.

(3)

(b) (i) Given that

$$7\sin^2\theta + \sin\theta\cos\theta = 6$$

show that

$$\tan^2\theta + \tan\theta - 6 = 0$$

(3)

(ii) Hence solve the equation $7\sin^2\theta + \sin\theta\cos\theta = 6$ in the interval $0° \le \theta \le 360°$, giving your answers to the nearest degree.

(4)

(Total 10 marks)

Q17.

(a) Solve the equation $\sec x = -5$, giving all values of x in radians to two decimal places in the interval $0 < x < 2\pi$.

(3)

(b) Show that the equation

$$\frac{\operatorname{cosec} x}{1+\operatorname{cosec} x}-\frac{\operatorname{cosec} x}{1-\operatorname{cosec} x}=50$$

can be written in the form

$$\sec^2 x = 25$$

(4)

(c) Hence, or otherwise, solve the equation

$$\frac{\operatorname{cosec} x}{1+\operatorname{cosec} x}-\frac{\operatorname{cosec} x}{1-\operatorname{cosec} x}=50$$

giving all values of x in radians to two decimal places in the interval $0 < x < 2\pi$.

(3)

(Total 10 marks)

Q18.

(a) (i) Given that $\tan 2x + \tan x = 0$, show that $\tan x = 0$ or $\tan^2 x = 3$.

(3)

(ii) Hence find all solutions of $\tan 2x + \tan x = 0$ in the interval $0° < x < 180°$.

(1)

(b) (i) Given that $\cos x \neq 0$, show that the equation

$$\sin 2x = \cos x \cos 2x$$

can be written in the form

$$2\sin^2 x + 2\sin x - 1 = 0$$

(3)

(ii) Show that all solutions of the equation $2\sin^2 x + 2\sin x - 1 = 0$ are given by

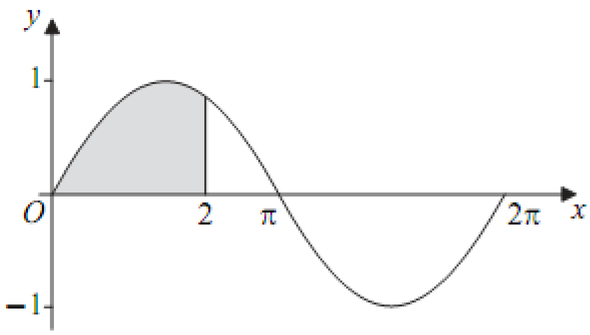
$\sin x = \dfrac{\sqrt{3}-1}{p}$, where p is an integer.

(3)

(Total 10 marks)

Q19.

A curve C, defined for $0 \le x \le 2\pi$ by the equation $y = \sin x$, where x is in radians, is sketched below. The region bounded by the curve C, the x-axis from 0 to 2 and the line $x = 2$ is shaded.



(a) The area of the shaded region is given by $\int_0^2 \sin x \, dx$, where x is in radians.

Use the trapezium rule with five ordinates (four strips) to find an approximate value for the area of the shaded region, giving your answer to three significant figures.

(4)

(b) Describe the geometrical transformation that maps the graph of $y = \sin x$ onto the graph of $y = 2 \sin x$.

(2)

(c) Use a trigonometrical identity to solve the equation

$$2 \sin x = \cos x$$

in the interval $0 \le x \le 2\pi$, giving your solutions in radians to three significant figures.

(4)

(Total 10 marks)

Q20.

(a) (i) Solve the equation $\operatorname{cosec}\theta = -4$ for $0° < \theta < 360°$, giving your answers to the nearest 0.1°.

(2)

(ii) Solve the equation

$$2 \cot^2(2x + 30°) = 2 - 7 \operatorname{cosec}(2x + 30°)$$

for $0° < x < 180°$, giving your answers to the nearest 0.1°.

(6)

(b) Describe a sequence of two geometrical transformations that maps the graph of $y = \operatorname{cosec} x$ onto the graph of $y = \operatorname{cosec}(2x + 30°)$.

(4)

(Total 12 marks)

Q21.

(a) Solve the equation $\tan(x + 52°) = \tan 22°$, giving the values of x in the interval

$0° \leq x \leq 360°$.

(3)

(b) (i) Show that the equation

$$3\tan\theta = \frac{8}{\sin\theta}$$

can be written as

$$3\cos^2\theta + 8\cos\theta - 3 = 0$$

(3)

(ii) Find the value of $\cos\theta$ that satisfies the equation

$$3\cos^2\theta + 8\cos\theta - 3 = 0$$

(2)

(iii) Hence solve the equation

$$3\tan 2x = \frac{8}{\sin 2x}$$

giving all values of x to the nearest degree in the interval $0° \leq x \leq 180°$.

(4)

(Total 12 marks)

Q22.

(a) Solve the equation

$$\operatorname{cosec} x = 3$$

giving all values of x in radians to two decimal places, in the interval $0 \leq x \leq 2\pi$.

(2)

(b) By using a suitable trigonometric identity, solve the equation

$$\cot^2 x = 11 - \operatorname{cosec} x$$

giving all values of x in radians to two decimal places, in the interval $0 \leq x \leq 2\pi$.

(6)

(Total 8 marks)

Q23.

(a) Express $\cos x + 3\sin x$ in the form $R\cos(x - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. Give your value of α, in radians, to three decimal places.

(3)

(b) (i) Hence write down the minimum value of $\cos x + 3 \sin x$.

(1)

(ii) Find the value of x in the interval $0 \leq x \leq 2\pi$ at which this minimum occurs, giving your answer, in radians, to three decimal places.

(2)

(c) Solve the equation $\cos x + 3 \sin x = 2$ in the interval $0 \leq x \leq 2\pi$, giving all solutions, in radians, to three decimal places.

(4)

(Total 10 marks)

Q24.

(a) Sketch the graph of $y = \cos x$ in the interval $0 \leq x \leq 2\pi$. State the values of the intercepts with the coordinate axes.

(2)

(b) (i) Given that

$$\sin^2 \theta = \cos \theta(2 - \cos \theta)$$

prove that $\cos \theta = \frac{1}{2}$.

(2)

(ii) Hence solve the equation

$$\sin^2 2x = \cos 2x(2 - \cos 2x)$$

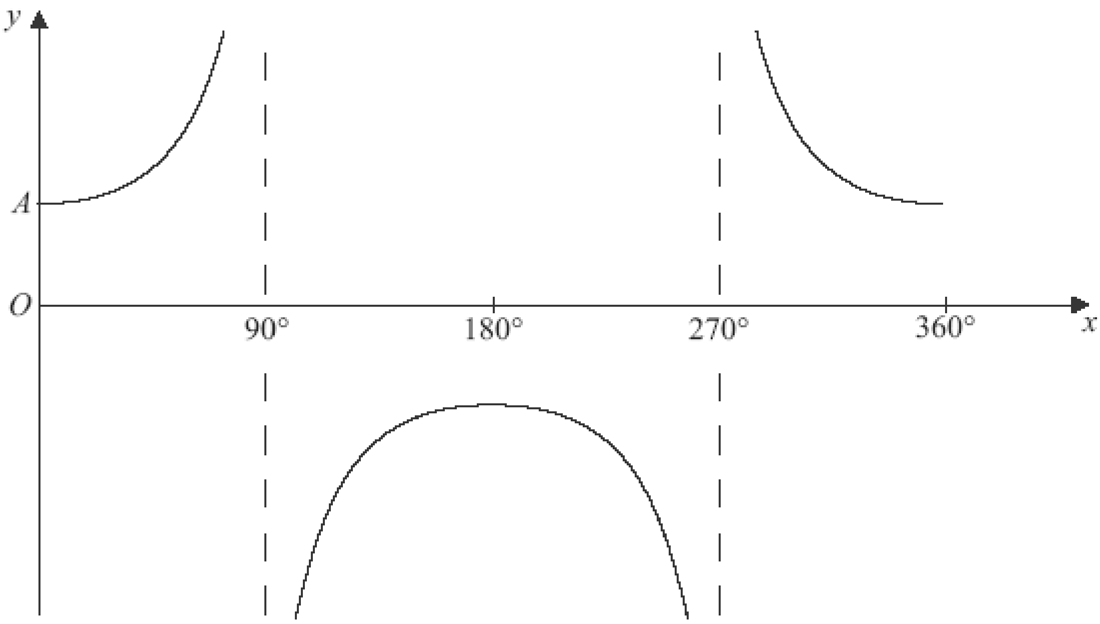
in the interval $0 \leq x \leq \pi$, giving your answers in radians to three significant figures.

(4)

(Total 8 marks)

Q25.

(a) The diagram shows the graph of $y = \sec x$ for $0° \leq x \leq 360°$.



The point A on the curve is where $x = 0$. State the y-coordinate of A.

(1)

(b) Solve the equation $\sec x = 2$, giving all values of x in degrees in the interval $0° \le x \le 360°$.

(2)

(Total 3 marks)

Q26.

(a) Find $\frac{dy}{dx}$ when:

(i) $y = \ln(5x - 2)$;

(2)

(ii) $y = \sin 2x$.

(2)

(b) The functions f and g are defined with their respective domains by

f(x) = ln(5x – 2), for real values of x such that $x \ge \frac{1}{2}$

g(x) = sin 2x, for real values of x in the interval $-\frac{\pi}{4} \le x \le \frac{\pi}{4}$

(i) Find the range of f.

(2)

(ii) Find an expression for gf(x).

(1)

(iii) Solve the equation $gf(x) = 0$.

(3)

(iv) The inverse of g is g^{-1}. Find $g^{-1}(x)$.

(2)
(Total 12 marks)

Q27.

(a) Show that the equation

$$10 \operatorname{cosec}^2 x = 16 - 11 \cot x$$

can be written in the form

$$10 \cot^2 x + 11 \cot x - 6 = 0$$

(1)

(b) Hence, given that $10 \operatorname{cosec}^2 x = 16 - 11 \cot x$, find the possible values of $\tan x$.

(4)
(Total 5 marks)

Q28.

(a) (i) Show that the equation $3 \cos 2x + 2 \sin x + 1 = 0$ can be written in the form

$$3 \sin^2 x - \sin x - 2 = 0$$

(3)

(ii) Hence, given that $3 \cos 2x + 2 \sin x + 1 = 0$, find the possible values of $\sin x$.

(2)

(b) (i) Express $3 \cos 2x + 2 \sin 2x$ in the form $R \cos (2x - \alpha)$, where $R > 0$ and $0° < \alpha < 90°$, giving α to the nearest 0.1°.

(3)

(ii) Hence solve the equation

$$3 \cos 2x + 2 \sin 2x + 1 = 0$$

for all solutions in the interval $0° < x < 180°$, giving x to the nearest 0.1°.

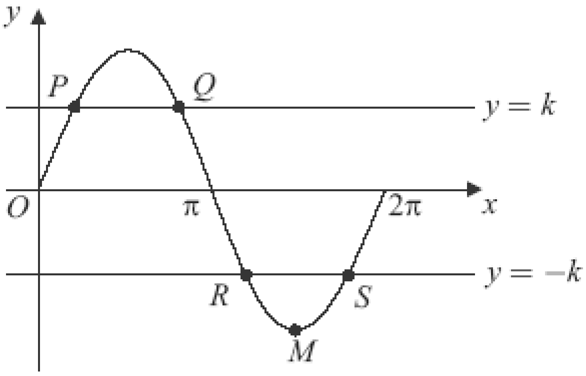
(3)
(Total 11 marks)

Q29.

(a) Solve the equation $\sin x = 0.8$ in the interval $0 \leq x \leq 2\pi$, giving your answers in radians to three significant figures.

(3)

(b) The diagram shows the graph of the curve $y = \sin x$, $0 \leq x \leq 2\pi$ and the lines $y = k$ and $y = -k$.



The line $y = k$ intersects the curve at the points P and Q, and the line $y = -k$ intersects the curve at the points R and S.

The point M is the minimum point of the curve.

(i) Write down the coordinates of the point M.

(2)

(ii) The x-coordinate of P is α.

Write down the x-coordinate of the point Q in terms of π and α.

(1)

(iii) Find the length of RS in terms of π and α, giving your answer in its simplest form.

(2)

(c) Sketch the graph of $y = \sin 2x$ for $0 \leq x \leq 2\pi$, indicating the coordinates of points where the graph intersects the x-axis and the coordinates of any maximum points.

(5)

(Total 13 marks)

Q30.

(a) Solve the equation $\sec x = \frac{3}{2}$, giving all values of x to the nearest degree in the interval $0° < x < 360°$.

(2)

(b) By using a suitable trigonometrical identity, solve the equation

$$2 \tan^2 x = 10 - 5 \sec x$$

giving all values of x to the nearest degree in the interval $0° < x < 360°$.

(6)

(Total 8 marks)

Q31.

(a) Express sin $2x$ in terms of sin x and cos x.

(1)

(b) Solve the equation

$$5 \sin 2x + 3 \cos x = 0$$

giving all solutions in the interval $0° \leq x \leq 360°$ to the nearest 0.1°, where appropriate.

(4)

(c) Given that $\sin 2x + \cos 2x = 1 + \sin x$ and $\sin x \neq 0$, show that $2(\cos x - \sin x) = 1$.

(4)

(Total 9 marks)

Q32.

(a) Given that $\dfrac{\sin\theta - \cos\theta}{\cos\theta} = 4$, prove that $\tan\theta = 5$.

(2)

(b) (i) Use an appropriate identity to show that the equation

$$2 \cos^2 x - \sin x = 1$$

can be written as

$$2 \sin^2 x + \sin x - 1 = 0$$

(2)

(ii) Hence solve the equation

$$2 \cos^2 x - \sin x = 1$$

giving all solutions in the interval $0° \leq x \leq 360°$.

(5)

(Total 9 marks)

Q33.

(a) Solve the equation $\tan x = -\dfrac{1}{3}$, giving all the values of x in the interval $0 < x < 2\pi$ in radians to two decimal places.

(3)

(b) Show that the equation

$$3 \sec^2 x = 5(\tan x + 1)$$

can be written in the form $3 \tan^2 x - 5 \tan x - 2 = 0$.

(1)

(c) Hence, or otherwise, solve the equation

$$3 \sec^2 x = 5(\tan x + 1)$$

giving all the values of x in the interval $0 < x < 2\pi$ in radians to two decimal places.

(4)

(Total 8 marks)

Q34.

(a) (i) Show that the equation $3 \cos 2x + 7 \cos x + 5 = 0$ can be written in the form $a \cos^2 x + b \cos x + c = 0$, where a, b and c are integers.

(3)

(ii) Hence find the possible values of $\cos x$.

(2)

(b) (i) Express $7 \sin \theta + 3 \cos \theta$ in the form $R \sin(\theta + \alpha)$, where $R > 0$ and α is an acute angle. Give your value of α to the nearest 0.1°.

(3)

(ii) Hence solve the equation $7 \sin \theta + 3 \cos \theta = 4$ for all solutions in the interval $0° \leq \theta \leq 360°$, giving θ to the nearest 0.1°.

(3)

(c) (i) Given that β is an acute angle and that $\tan \beta = 2\sqrt{2}$, show that $\cos \beta = \dfrac{1}{3}$.

(2)

(ii) Hence show that $\sin 2\beta = p\sqrt{2}$, where p is a rational number.

(2)

(Total 15 marks)

Q35.

(a) Given that

$$\frac{3 + \sin^2 \theta}{\cos \theta - 2} = 3 \cos \theta$$

show that

$$\cos \theta = -\frac{1}{2}$$

(4)

(b) Hence solve the equation

$$\frac{3+\sin^2 3x}{\cos 3x - 2} = 3\cos 3x$$

giving all solutions in degrees in the interval $0° < x < 180°$.

(4)

(Total 8 marks)